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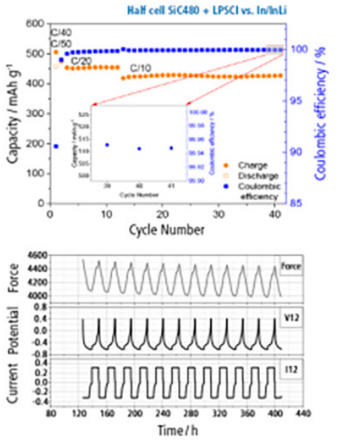
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Improving Low-Load Efficiency of Marine SOFC Powertrains by Means of Efficient Module Hot Standby Operation

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Solid oxide fuel cell (SOFC) powertrains can be considered as a promising solution for future zero-emission shipping, yet the decrease in electrical efficiency at low load remains a barrier for full-scale marine adoption. This work provides the first comprehensive quantification of hot standby power demands for marine SOFC modules and evaluates its impact on the multi-modular powertrain efficiency. A thermodynamic process system model of a 112 kW_{el} rated SOFC module was utilized, including anode recirculation, autothermal pre-reforming in hot standby, heat losses and off-design heat exchanger behavior. Assessing five hot standby heat integration variants, combined fuel-equivalent standby demands ranging from 5.2 to 15.3 kW_{chem} per module were obtained. The highest-performing configurations are those employing cathode off-gas recirculation, motivating their integration in a multi-modular marine SOFC concept. At the multi-module powertrain level, non-uniform operation strategies, in which individual modules either operate close to their maximum-efficiency point or remain in hot standby, can significantly increase low-load efficiency. At a typical cruise-ship operating point near 15% rated power, efficiencies of up to 55% (LHV to AC) are achieved, compared to only 26% for uniform operation of all modules. The results highlight the importance of incorporating hot standby strategies into future marine SOFC system design.
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Supplementary material for this article is available online

Solid Oxide Fuel Cells (SOFCs) have emerged as a promising alternative to conventional internal combustion engine (ICE)-based propulsion systems for seagoing vessels, as they offer substantially higher fuel-to-electricity conversion efficiencies and lower carbon dioxide (CO_2) emissions. Compared with liquefied natural gas (LNG)-fueled ICEs, SOFCs operating on the same fuel are able to almost completely eliminate methane slip while achieving notable reductions in other non-greenhouse gas emissions such as nitrogen oxides (NO_x).[1] A further advantage lies in their fuel flexibility, as SOFCs can operate on a range of future marine fuel candidates, including LNG, methanol and ammonia.[2]

The integration of SOFCs into shipboard power systems also provides opportunities for waste heat recovery. The thermal output, released with the exhaust gas of the SOFC system at temperatures around 200 °C, can be fed into the ship's saturated steam and hot water grids. In this way, the overall efficiency of the installation is increased by combining electrical power generation with useful thermal energy supply.[3]

A well-documented drawback of SOFC systems, however, is their significant efficiency penalty at low part-load operation. As the load decreases, stack heat dissipation and overall mass flows decrease, and would result in lower stack temperatures and lower effectiveness of system-level heat integration. To compensate this, additional heat is required, typically achieved by means of a lowered fuel utilization or excess fuel to the afterburner to maintain operating temperatures, leading to steep declines in electrical efficiency. This trend is illustrated in Fig. 1a, which compiles field data from stationary systems[1] (blue circles) alongside a functional demonstrator test emulating maritime powertrain conditions executed as part of the Nautilus project by the authors[4] (black squares). In this publication, the electrical efficiency of an SOFC system is consistently defined as the ratio of AC electrical output power to the overall LHV fuel input:

$$\eta_{\mathrm{el,LHV,AC}} = \frac{P_{\mathrm{sys,AC,out}}}{\dot{m}_{\mathrm{fuel,in}} \cdot LHV_{\mathrm{fuel}}}. \quad [1]$$

Further on, all electrical powers in this manuscript are considered AC if not stated otherwise. Both experimental datasets, as well as steady-state thermodynamic simulations of an SOFC system model presented in this publication, show electrical efficiencies dropping well below 40% at operating points below 20% of the rated system power.

Modern cruise vessels typically employ an integrated electric propulsion (IEP) concept, where ICE gensets supply power to a common electrical grid that serves propulsion motors, hotel loads, and auxiliary systems. Such architectures are favored for their operational and design flexibility as well as improved overall efficiency.[5] A normalized load demand distribution from such a multi-MW IEP cruise ship on a 7-day cruise in the Mediterranean obtained in the Nautilus project is shown in Fig. 1b. Significant periods of low-load operation are visible, with a peak around 15% of maximum demand during port stays, depicted as a dashed vertical black line.

In the context of maritime SOFC integration, multiple levels of SOFC-ICE hybridization have been investigated by van Veldhuizen et al.[6] In layouts with small SOFC shares, SOFCs operate primarily as base-load generators to cover continuous hotel and auxiliary demands, while ICEs handle load fluctuations, minimizing SOFC exposure to inefficient low-load operation. In contrast, the more ambitious full SOFC ship installation offers the highest theoretical efficiency benefits and lowest CO_2 and non-CO_2 emissions. Apart from the present limitations due to low SOFC system power densities, the challenge in a full SOFC powertrain scenario lies in maintaining acceptable efficiencies under prolonged low-load conditions. Uniformly distributing the reduced power demand across all installed modular SOFC systems is not viable, as it would increase fuel consumption and NO_x emissions,[4] particularly in emission-controlled harbor areas. Conversely, shutting down modules completely is operationally impractical, since the time required for a full cool-down and subsequent reheating of current SOFC stacks is equal or even longer than a typical 10 to12 h port stay.

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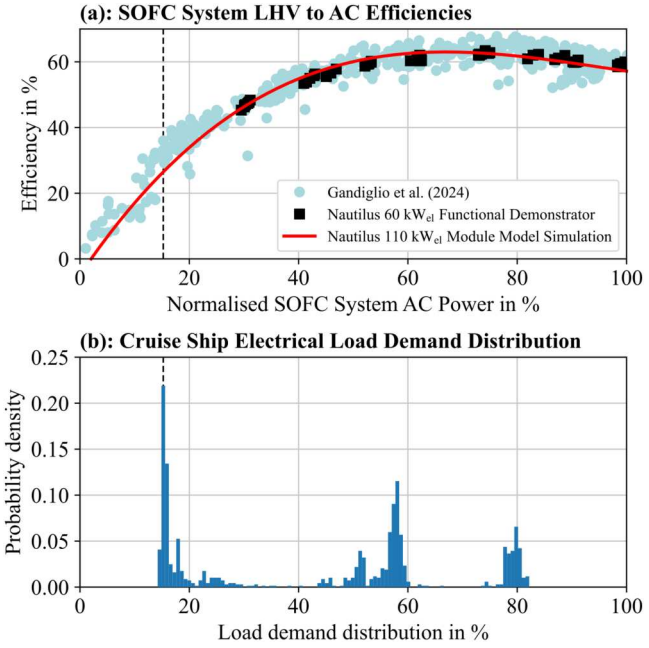


Figure 1. (a) LHV to AC Efficiency of both experimental and simulated SOFC systems,[1,4] with kind permission from Gandiglio et al. (b) Load demand distribution including hotel load, auxiliary and propulsive power of a cruise ship on a 7-day mission.

An alternative strategy is a non-uniform operation, whereby only the necessary number of modules operates near their most efficient point, while the remaining modules are put in hot standby mode. This approach ensures rapid availability of the full rated power while avoiding low part-load operation of any module. Previous studies in stationary SOFC-battery systems have highlighted hot standby as a promising strategy to reconcile operational flexibility with durability. In particular, Nousch et al. reported that it enables stable high-temperature operation with reduced need for auxiliary storage, though at the expense of higher fuel consumption.[7] The exact standby power demands, both in terms of fuel and electricity, remain poorly quantified, and their impact on overall system efficiency under realistic boundary conditions has not yet been systematically assessed.

To the authors' knowledge, this specific research question has not been comprehensively addressed in the literature. The present work therefore aims to close this gap by:

1. Quantifying the theoretical powertrain efficiency potential of modular non-uniform operation with hot standby, as a function of power demand for hot standby.
2. Identifying suitable hot standby operating modes and layouts through thermodynamic steady-state simulations, including additional balance-of-plant (BoP) requirements and operational challenges.
3. Providing realistic estimates of hot standby demands and their implications for maritime SOFC powertrain feasibility.

Methodology

The methodology applied in this work is structured as follows. First, the overall modular maritime SOFC powertrain design is defined, including the SOFC module as the smallest controllable unit and its hierarchical integration into larger units and ship-level installations. The modeling framework is then introduced, detailing the SOFC module assumptions, the nominal operating characteristics and the validation basis. A suitable key performance indicator, the overall powertrain or unit efficiency, is established, and its theoretical sensitivity to hot standby power demand is derived. Subsequently, the requirements and boundary assumptions for hot standby operation are discussed. Finally, five distinct module layouts are presented, differing in their hot standby heat supply and integration concepts, which form the basis for the comparative analysis.

SOFC powertrain modular design.—The overall modular design philosophy of the investigated powertrain was established in the Nautilus project and is illustrated in Fig. 2. At the core of the architecture is the SOFC module, comprising six stacks with a rated direct current (DC) output of 22.5 $kW_{el,DC}$ at beginning of life (BoL). The stacks are integrated with common hot BoP components in a gas-tight compartment, including fuel and air supply, anode off-gas recirculation (AOGR) for steam supply to the pre-reformer and overall high fuel utilization, excess fuel combustion, and internal heat integration by means of a heat exchanger network explained in detail by Pina et al..[8] This configuration results in a module rated at 112 $kW_{el,AC}$ throughout the entire system lifetime, as already stated in first system concepting.[3] Determining the module size was driven by ship integrability and spacing dimensions. The module thus represents the smallest individually controllable unit of the powertrain, as all stacks within it are electrically actuated uniformly and supplied by a common fuel input.

At the next integration level, nine modules are combined into a unit, corresponding to one so-called fuel cell room on the ship. Such a unit therefore reaches an installed rated capacity of 1 $MW_{el,AC}$. In this arrangement, all modules share common cold BoP components, such as fuel desulphurization, air filtration and desalination, and exhaust gas heat recovery. Crucially, the individual modules within a unit can be operated independently, which is essential for the proposed hot standby operation strategy.

At the ship level, multiple units can be installed to achieve the desired total installed electrical power, depending on vessel size and mission profile. Additional battery storage may be integrated to provide peak-shaving capability and to support transient load demand.

SOFC module layout.—The investigated module was designed in detail as part of a Nautilus project deliverable as a conventional methane-fueled SOFC system with pre-reforming and AOGR designed for overall high electrical efficiency and ship integration constraints, as depicted in Fig. 3. Six stacks are arranged in a common insulated hot box and their DC output is converted by an inverter and fed to the common busbar, from which BoP consumers such as blowers and, if required, auxiliary heaters are supplied. A high-temperature anode recirculation blower is positioned in close proximity to the stacks. Fuel pre-reforming is conducted in an adiabatic pre-reformer, which is integrated between heat exchangers supplied by the afterburner exhaust. The heat exchangers are configured in parallel to provide heating of both anode and cathode streams, with mass flow distribution regulated by downstream valves.

As an optional design feature, a cathode off-gas recirculation (COGR) loop is considered. In this configuration, a fraction of the cathode exhaust is returned directly to the cathode inlet via a high-temperature air blower. While simulations indicate only a limited effect on electrical efficiency for nominal methane-fueled operation, this option provides a significant lever for enhanced heat utilization, thereby improving the overall thermal efficiency of the module.[3]

The simulated performance of this module layout, summarized in Table I, provides the rated power (100%) and the maximum efficiency (62%) operating points. Additional process simulation data explaining the operating conditions at part load and the resulting efficiency profile can be found as Supplementary Material. Although this specific configuration has not yet been experimentally validated, the calculated electrical efficiency as a function of load closely matches experimental results obtained from commercial SOFC systems with anode recirculation reported in literature[1,4] (see Fig. 1a).

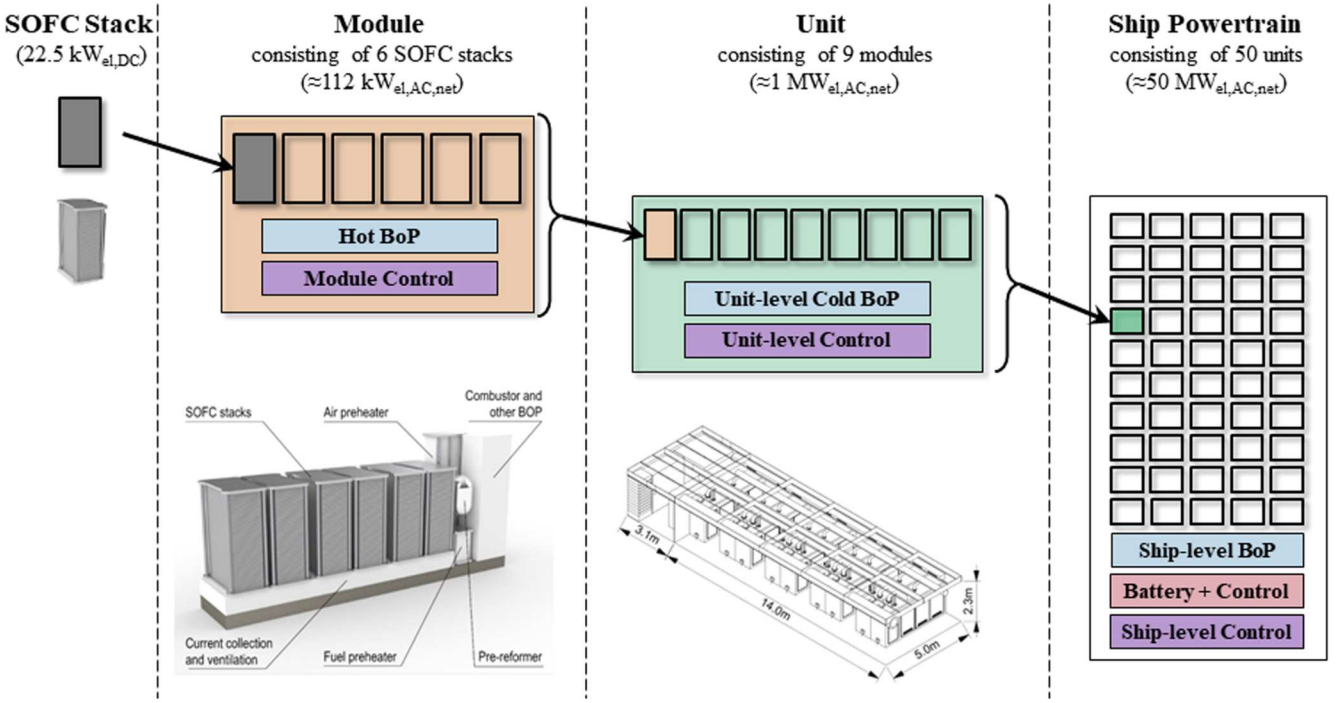


Figure 2. Modular design philosophy and CAD drawings from powertrain layout in the Nautilus project, consisting of SOFC stack, module, unit and overall ship integration.

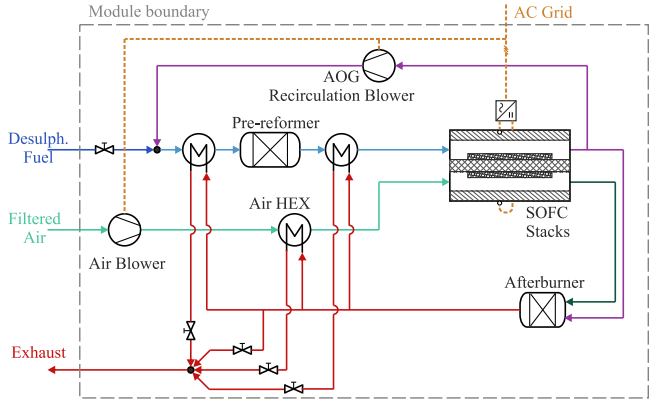


Figure 3. Module process flow diagram for nominal operation.

Impact of module hot standby power demand on unit's part load efficiency.—The first step of the analysis is to assess the theoretical benefit of non-uniform operation in a modular SOFC powertrain, in which a portion of the installed modules is kept in a hot standby state while the remaining ones operate actively. A unit consists of N identical modules, each of which can be either active or in hot standby, resulting in a number of active modules N_{act}, and standby modules $N - N_{act}$. The electrical efficiency of such a unit, accounting for both fuel and electrical power consumption of the hot standby modules, can be expressed as

$$\eta^{unit}_{el,LHV,AC} = \frac{\left|\sum_{k=1}^{N_{act}} P^{mod}_{el,act,k}\right| - \sum_{k=1}^{N-N_{act}} P^{mod}_{el,standby,k}}{LHV_{fuel}\cdot\left(\sum_{k=1}^{N_{act}} \dot{m}_{fuel,act,k} + \sum_{k=1}^{N-N_{act}} \dot{m}_{fuel,standby,k}\right)}, \quad [2]$$

where LHV_{fuel} denotes the lower heating value of the fuel, set to 50.14 MJ kg^{-1} for pure methane in this study. The default configuration comprises $N = 9$ modules, corresponding to the unit layout described in Fig. 2. The effect of varying the total number of modules is investigated later in the sensitivity study.

In a first theoretical assessment, the influence of hot standby fuel and electrical power demands on the overall unit efficiency is analyzed independently to obtain a general cause-and-effect relationship, even though both contributions occur simultaneously in practical operation. Figure 4 compares the electrical unit efficiency of conventional uniform operation, where all nine modules operate homogeneously (shown in red), with non-uniform operation, where only a subset of modules remains active while the others are kept in hot standby. The analysis focuses on part-load conditions below the maximum efficiency point of 62% rated AC power.

In Fig. 4, all active modules are assumed to operate at their individual maximum efficiency operating point, so that only $N = 9$ discrete unit output powers can be achieved below the maximum efficiency operating point. These operating points are marked by circles in Fig. 4, and the trend between them is indicated by dashed fit lines. At the rightmost edge of each plot, where all nine modules are active, the efficiency curves of uniform and non-uniform operation converge. As load decreases, additional modules are successively switched into hot standby.

Figure 4a illustrates the case where the hot standby demand is purely in the form of fuel consumption, $LHV_{fuel}\cdot\dot{m}_{fuel,standby}$, varied from 0 kW$_{chem}$ up to 30 kW$_{chem}$ per 112 kW$_{el,rated}$ module, while electrical power demand is neglected. This range corresponds to 0 to15% of the equivalent fuel demand during rated operation. For the idealized case of zero standby fuel consumption, the overall unit efficiency theoretically remains constant at the maximum module efficiency across all discrete part-load points, as the standby modules do not require additional energy. With increasing standby fuel demand, the unit efficiency gradually decreases until, at approximately 30 kW$_{chem}$ per module, the benefit over uniform operation becomes negligible. Because total power output remains constant, the operating points corresponding to an identical number of active modules align vertically in the plot, as for the example of 21% of rated power, where three modules are active and six are in hot standby.

Figure 4b presents the complementary theoretical scenario where all hot standby power is supplied electrically. The electrical power

Table I. Module performance specification at rated power and max. efficiency operating points as well as in hot standby.

Variable		Unit	Rated power	Max. efficiency	Hot standby
SOFC DC Power $P_{\mathrm{SOFC,DC}}$		kW	125.4	76.3	0
Module AC Output Power $P_{\mathrm{el}}^{\mathrm{mod}}$		kW	112.1	69.5	0
Module LHV to AC efficiency $\eta_{\mathrm{el,LHV,AC}}^{\mathrm{mod}}$		%	57.8	63.4	—
Fuel supply characteristics					
Required fuel mass flow $\dot{m}_{\mathrm{fuel,in}}$		g/s	3.87	2.32	0.04
Mass flow in pre-reformer $\dot{m}_{\mathrm{ref}}$		g/s	30.1	16.6	8.36
Stack fuel utilization FU_{stack}		—	0.75	0.75	—
Anode recirculation ratio RR_{AOGR}		—	0.6	0.6	0.95
O/C ratio		—	2.1	2.1	2.1
Pre-reformer outlet molar fractions	H_2 $x_{\mathrm{H_2,ref,out}}$	%	24.8	24.8	17.3
	CH_4 $x_{\mathrm{CH_4,ref,out}}$	%	12.3	12.3	< 0.1
	CO_2 $x_{\mathrm{CO_2,ref,out}}$	%	25.3	25.3	6.2
	CO $x_{\mathrm{CO,ref,out}}$	%	4.0	4.0	7.7
	H_2O $x_{\mathrm{H_2O,ref,out}}$	%	33.6	33.6	10.7
	N_2 $x_{\mathrm{N_2,ref,out}}$	%	0	0	58.0
Air supply characteristics					
Cathode inlet mass flow $\dot{m}_{\mathrm{cath,in}}$		g/s	367	122	40
Air inlet mass flow $\dot{m}_{\mathrm{air,in}}$		g/s	367	122	no COGR: 40 with COGR: 6
Cathode recirculation ratio RR_{COGR}		—	0	0	no COGR: 0 with COGR: 0.85
Air mass flow to pre-reformer $\dot{m}_{\mathrm{air,ATR}}$		g/s	—	—	0.38

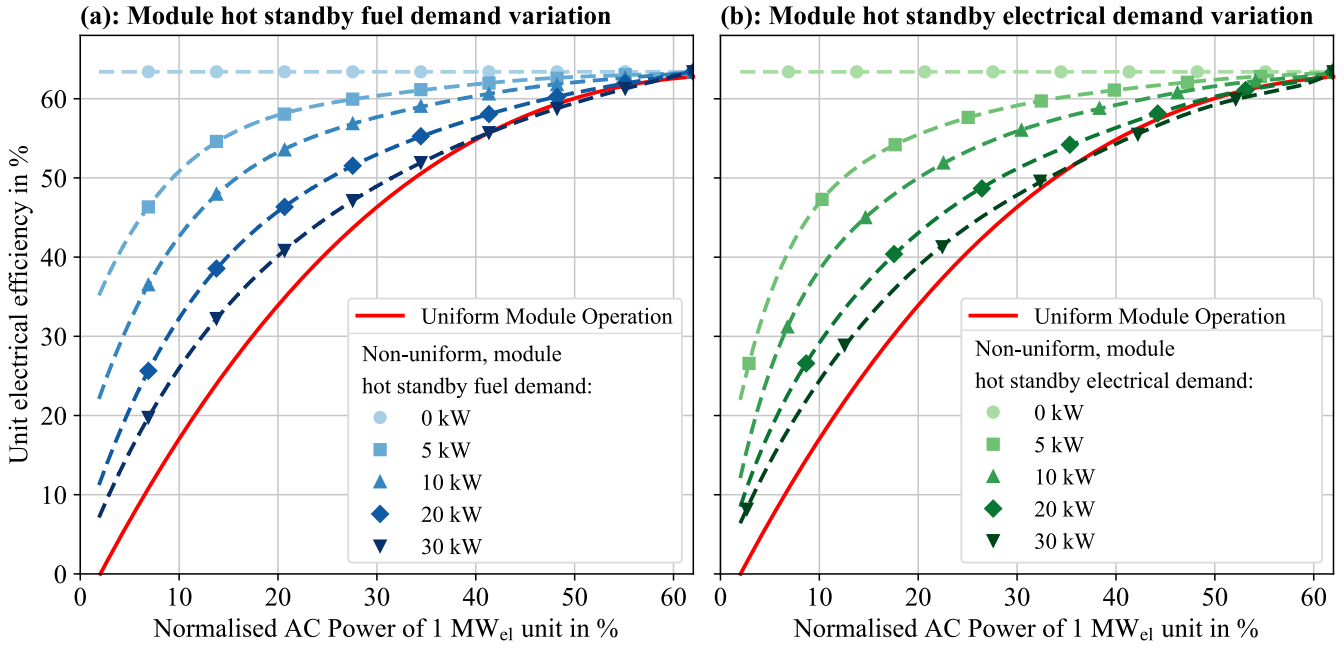


Figure 4. Comparison of theoretical unit efficiency at low part load power demand, with all 9 modules of unit operating uniformly (red lines) and non-uniform operation with variation of hot standby module (a) fuel and (b) electrical power demand.

demand per standby module $P_{\mathrm{el,standby}}^{\mathrm{mod}}$, is varied from 0 $\mathrm{kW_{el}}$ up to 30 $\mathrm{kW_{el}}$. The resulting efficiency reduction is more pronounced than in the fuel-based case, as higher standby electrical power must be provided by the active modules themselves, effectively lowering the net unit power output. Consequently, the discrete operating points shift toward lower total output power. In other words, for a fixed required unit power, higher standby electrical demands require a greater number of modules remaining active only to cover the internal power consumption.

This theoretical assessment of the limiting cases of purely fuel- and purely electricity-driven standby operation demonstrates the potential of a modular SOFC powertrain to maintain high electrical efficiencies even at low overall power demand. However, for such a system to yield a competitive advantage, the total standby power demand per 112 $\mathrm{kW_{el}}$-rated module is proposed to remain below or in the range of 10 kW, corresponding to roughly 9% of the rated electrical output power or 5% of the rated fuel input. These findings motivate the detailed investigation of feasible and efficient hot standby operating conditions presented in the following sections.

***Hot standby operating conditions.*—**In the hot standby state, the SOFC module does not supply electrical power to the ship's grid but

remains in a thermally active condition close to its nominal operating temperature. The objective of this operating mode is to minimize the module's overall energy demand while ensuring that it can promptly resume full-power operation when required.

Apart from maintaining the overall temperature level, the catalytic surfaces within the anode and pre-reformer must remain in a reducing atmosphere to avoid oxidation of the nickel-based catalysts. Furthermore, the oxygen-to-carbon (O/C) ratio in the supplied gases should be kept above a temperature-dependent threshold to prevent carbon formation.

Different hot standby operating modes can be categorized according to the state of the SOFC stacks, fuel supply, and thermal integration strategy:

State of the SOFC stacks:

- *Open-circuit voltage (OCV) mode:* The SOFC operates without generating electrical power; any electrical demand from the BoP components is provided by the grid or, in a modular unit, by other active modules.
- *Net-zero power mode:* The stacks operate at very low current densities, producing just enough electrical power to supply their own BoP components, thereby avoiding the need for external electrical input.

State of the fuel supply:

- *Protection-gas mode:* Feeding an inert or forming gas mixture instead of fuel can maintain a reducing atmosphere. However, implementation in maritime applications would require additional onboard inert-gas infrastructure and storage capacity, similar to systems already employed on certain vessels handling low-flashpoint fuels, which is feasible, yet would increase overall system complexity and integration effort.
- *Continued fuel supply mode:* The preferred option for shipboard systems, where natural gas is supplied at significantly reduced mass flow rates to provide the necessary hydrogen and carbon monoxide for maintaining a reducing atmosphere in the anode loop.

State of thermal integration:

- *Afterburner heat recovery:* High-temperature heat from the afterburner is transferred via heat exchangers to the SOFC stacks to compensate for system heat losses.
- *Auxiliary electrical or fuel-based heating:* Additional electrical heaters or burners can be placed upstream of the cathode to supply heat directly.
- *Autothermal reforming (ATR):* A small controlled amount of air is introduced to the pre-reformer, enabling partial oxidation of methane in parallel to steam reforming to sustain a target pre-reformer outlet temperature while converting a large fraction of methane before entering the stack.
- *Cathode off-gas recirculation (COGR):* A portion of the cathode exhaust is fed back to the inlet using a high-temperature air blower, keeping a hot airflow around the stack assembly. The approach reduces the need for external air supply while maintaining adequate cathode flow through the stacks.

In this study, and considering the boundary conditions of maritime applications, only hot standby operation modes are analyzed that (i) maintain a continuous natural gas supply, as no alternative reducing gases are available, and (ii) operate at open-circuit voltage (OCV). This is justified by the modular drivetrain concept, in which electrical energy can be efficiently provided by other active modules operating at high conversion efficiencies.

Modeling framework and assumptions.—The following section describes the modeling approach used to assess realistic power demands for the investigated SOFC module under hot standby conditions. All simulations were conducted using the Modelica-based TEMPEST framework, developed at DLR specifically for modeling and validating multi-stack SOFC assemblies and their associated BoP components. As the focus of this work lies on thermal management rather than electrochemical performance, only the OCV condition as well as chemical reactions at the anode catalyst of the SOFC are considered. Special emphasis is placed on modeling heat transfer in system off-design operation, heat losses to the ambient, reaction equilibria in the pre-reformer, as well as pressure losses and their resulting BoP power demands. The underlying component models and governing equations have been described in detail in previous publications by the authors.[9,10]

Pre-reformer.—A lumped Gibbs reactor model is employed to capture the thermal behavior and methane conversion of the pre-reformer in autothermal reforming mode in hot standby, considering the following six chemical reactions:

$CH_4 + 2\,O_2 \rightleftharpoons CO_2 + 2\,H_2O$	Complete oxidation of methane
$CH_4 + 0.5\,O_2 \rightleftharpoons CO + 2\,H_2$	Partial oxidation of methane (POX)
$CH_4 + O_2 \rightleftharpoons CO_2 + 2\,H_2$	Incomplete oxidation of methane
$CO + H_2O \rightleftharpoons CO_2 + H_2$	Water – gas shift (WGS) reaction
$CH_4 + H_2O \rightleftharpoons CO + 3\,H_2$	Steam reforming of methane (MSR)
$CH_4 + CO_2 \rightleftharpoons 2\,CO + 2\,H_2$	Dry reforming of methane

[3]

Because of the low overall mass flows and the high catalytic surface availability in hot standby, thermodynamic equilibrium with respect to these reactions is assumed at the pre-reformer outlet. Carbon-forming reactions, such as the Boudouard reaction or methane cracking, are neglected, as the selected O/C ratio is controlled to a value of 2.1, sufficiently high to prevent solid carbon deposition at the targeted pre-reformer temperature above 600 °C. Validation of the equilibrium solver was carried out by comparing simulated methane conversion with the temperature-dependent data of Enger et al.,[11] resulting in deviations below 2% across the relevant temperature and methane-to-oxygen ratio range required in this study.

SOFC stacks.—Whereas most literature focuses on spatially discretized cell or stack models to resolve electrochemical and thermal coupling effects, such detailed representation is unnecessary here, as no electrochemical reactions occur in OCV operation. Instead, a lumped SOFC model obtained from a discretized model by means of model reduction was utilized. Parameterization was done to depict the lumped behavior of a SolydEra G8-X9 stack that consists of an arrayed 3×3 arrangement of nine 80 cm^2 anode-supported single cells per layer.[12] Lumped electrochemical performance was taken from experimental testing of a SolydEra module in reversible system operation, consisting of four smaller 2×2 G8-X4 stacks with the identical cell materials.[13]

Within this model, complete methane conversion by means of MSR at the anode outlet is assumed, justified by (i) pre-reformer conversion levels exceeding 90% as defined in the next section and (ii) stack temperatures remaining above 600 °C in all simulated conditions. The water-gas shift reaction is assumed to reach equilibrium at the outlet. In the absence of electrochemical heat generation, the stack energy balance reduces to the reaction enthalpy of MSR and WGS, enthalpy flows of incoming and outgoing gases and the thermal losses to the surroundings, given by

$$\dot{Q}_{loss,stack}(T_{stack,avg}) = (h{\cdot}A)_{loss}{\cdot}(T_{stack,avg} - T_{amb}), \qquad [4]$$

where $(h{\cdot}A)_{loss}$ is the overall heat loss coefficient, tuned to achieve a nominal heat loss of 2.3 kW of the six-stack hot box assembly under rated conditions. BoP component losses are modeled analogously, with a combined heat loss of 0.7 kW in nominal operation. These values are based on detailed three-dimensional thermal steady-state simulations of the module installed as part of a 1 MW unit in a ship

engine room environment (ambient temperature 45 °C, insulation thickness 10 cm for the stack and hot BoP). The distribution of BoP heat losses among individual components is summarized in Table II.

Heat exchangers.—Accurate modeling of the heat exchangers is essential, as in hot standby operation they operate far from design conditions at significantly reduced mass flows. It is assumed that the heat exchangers are designed for rated conditions, while keeping pressure losses below 10 mbar in order to comply with the stack's differential pressure constraints to ambient. Their effectiveness under hot standby is calculated using a counter-flow NTU method. Rather than assuming a constant overall heat transfer coefficient, Nusselt-number correlations that account for variations in Reynolds numbers and thus mass flow and gas properties changes are applied. The approach and implementation are based on validated high-temperature gas–gas heat exchanger experiments with low pressure drops.[14,15]

Pressure losses and BoP power demand.—Design pressure drops at rated conditions are set to 10 mbar for all BoP components, and to 15 mbar and 20 mbar for the SOFC anode and cathode, respectively, in line with a previous design study.[3] Pressure losses in hot standby are estimated using a linear mass-flow dependence occurring at laminar flow:

$$\Delta p_{\mathrm{lam},i} = \epsilon \cdot k_{\mathrm{p},i} \cdot \dot{m}_i, \qquad [5]$$

where $k_{\mathrm{p},i}$ represents the pressure loss coefficient of component i, calibrated to match the design pressure drop at rated mass flow, while ϵ is an overall scaling factor (set to 1 as reference) later used in the sensitivity study. The electrical power required for the air blower and the anode and cathode recirculation blowers is determined from the respective pressure differences to be overcome. For the exhaust and heat recuperation system of the vessel downstream of the drivetrain unit, a design pressure drop of 25 mbar at rated mass flow is assumed. The mechanical and isentropic efficiencies of the blowers are listed in Table II.

Hot standby approach and thermal integration variants.—Table I summarizes the reference settings applied in this study for hot standby operation. The air and fuel mass flows to the pre-reformer during autothermal reforming are controlled such that the O/C ratio remains identical to the value during active operation (2.1) and the pre-reformer outlet temperature is maintained at approximately 700 °C to achieve a degree of methane conversion above 95%.

Table II. Module model reference assumptions.

Variable	Unit	Value
SOFC multi-stack assembly at rated power		
SOFC hot box heat loss $\dot{Q}_{\mathrm{loss,SOFC,rated}}$	kW	2.3
Anode flow pressure loss $\Delta p_{\mathrm{an,rated}}$	mbar	20
Cathode flow pressure loss $\Delta p_{\mathrm{cath,rated}}$	mbar	15
BoP components at rated power		
Sum of BoP heat loss $\sum_i \dot{Q}_{\mathrm{loss,BoP,rated},i}$	kW	0.7
Heat loss share of fuel heat exchangers $k_{\mathrm{Q,fuelHEX}}$	%	15 each
Heat loss share of air heat exchanger $k_{\mathrm{Q,airHEX}}$	%	30
Heat loss share of afterburner $k_{\mathrm{Q,Afterburner}}$	%	20
Heat loss share of recirculation blowers $k_{\mathrm{Q,bl,rec}}$	%	10 each
Component design pressure loss $\Delta p_{\mathrm{lam,BoP,rated},i}$	mbar	10
Downstream exhaust pressure loss $\Delta p_{\mathrm{exh,rated}}$	mbar	25
Blower isentropic efficiency $\eta_{\mathrm{bl,is}}$	%	70
Blower mechanic efficiency $\eta_{\mathrm{bl,mech}}$	%	80

The overall process flows are significantly reduced compared to nominal operation but remain in a range able to ensure stable thermal conditions and maintain adequate heat transfer. Specifically, the heat capacity flows through the anode and cathode should be sufficiently high to compensate for the heat losses of the SOFC stack assembly without allowing the average stack temperature to drop by more than 50 K. In addition, very low flow rates may promote flow maldistribution within the stack, potentially leading to increased local thermal gradients. As these effects depend on the employed stack and manifold design, a feasible minimum air flow threshold remains uncertain at system level. Thus, a reference hot standby air mass flow of 10% of the nominal value is chosen as an engineering estimate (40 g s^{-1}, see Table I), yet the parameter is treated as a sensitivity parameter and investigated in detail in the results section. For concepts implementing cathode off-gas recirculation, with a reference recirculation ratio of $RR_{\mathrm{COGR}} = 85\,\%$, the external inlet air mass flow can be further reduced to about 15% of the corresponding non-COGR configuration (6 g s^{-1} instead of 40 g s^{-1}, see Table I) while maintaining the same total cathode flow through the stack.

To minimize the required inlet fuel flow while maintaining reducing conditions, the anode off-gas recirculation ratio RR_{AOGR} is increased substantially compared to nominal operation. At a reference setting of 95%, the majority of reducing species are retained within the loop, while a small fraction is depleted to prevent accumulation of inert gases or contaminants. Under these conditions, the mass flow in the pre-reformer and anode channels corresponds to approximately 28% of that during rated operation, whereas the fresh fuel supply entering the loop can be reduced to only about 1% of the rated value (see Table I).

The reference cathode inlet temperature setpoint to be achieved by all heat integration variants introduced below is set to 700 °C, representing the average cell temperature for anode-supported cells in operation. The afterburner is operated at a temperature setpoint of 750 °C, unless stated otherwise. Additional fuel may be supplied to the afterburner if the remaining combustible species from the anode loop are insufficient to sustain this minimum temperature.

Thermal integration variants.—Based on the above reference settings, five different thermal integration variants, listed in Table III, were defined to evaluate the effectiveness of heat supply concepts during hot standby operation. A schematic overview of the module layout with the relevant auxiliary components is shown in Fig. 5. The main additional subsystems include: (i) a dedicated air supply line to the pre-reformer for autothermal operation, (ii) an auxiliary fuel line feeding the afterburner, and (iii) an optional cathode air preheating device, depicted as a black-box component that can be realized either as a fuel-fired burner (blue line) or as an electric heater connected to the AC grid (orange dashed line).

The individual configurations are characterized as follows:

- Variant 1: Utilizes the existing heat exchanger network for thermal management. Heat losses are compensated by supplying additional fuel to the afterburner, which serves as the only heat source. The required additional fuel flow is a free variable adjusted to

Table III. Investigated plant variants, differing in hot standby heat integration.

Layout variant	ATR	Extra fuel afterburner	Cathode pre-heating	Cathode off-gas recirculation
1	X	X		
2	X	X	El. Heater	
3	X	X	Burner	
4	X		El. Heater	X
5	X		Burner	X

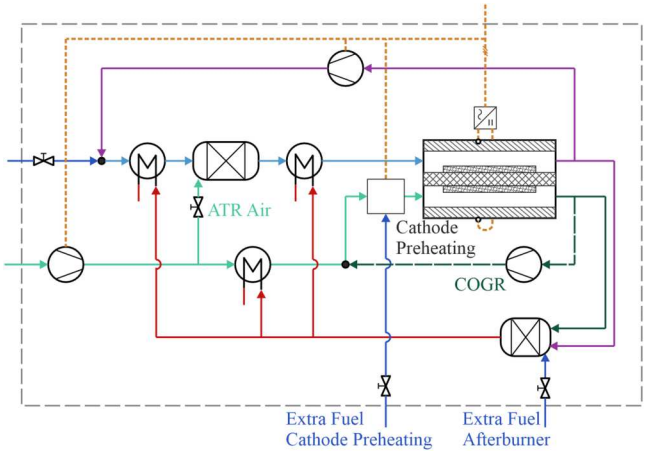


Figure 5. Process Flow Diagram depicting additional components used for hot standby operation, including air supply for ATR, extra fuel supply to the afterburner, either extra fuel or extra electrical power supply to cathode preheating, as well as cathode off-gas recirculation. Downstream exhaust piping is removed for visibility.

achieve the cathode inlet temperature setpoint. As the heat exchangers operate at low effectiveness under reduced mass flow conditions, the afterburner temperature is permitted to rise up to 900 °C.

- Variant 2: Employs an electric heater installed upstream of the air inlet of the stack hot box, a solution commonly applied in SOFC systems for start-up and controlled temperature maintenance.
- Variant 3: Replaces the electric heater with a fuel burner positioned upstream of the cathode, with the fuel flow to the burner being the free variable used to control the cathode inlet temperature. While this approach offers simple integration, the introduction of combustion products into the cathode stream can cause degradation of cathode materials.
- Variant 4: Like Variant 2, but combining the electric heater with COGR.
- Variant 5: Like Variant 3, but combining the burner with COGR, introducing a more pronounced risk of cathode degradation due to partial recirculation of combustion products within the cathode loop.

Results and Discussion

Hot standby fuel and electrical power demand at reference conditions.—A comparison of the fuel and electrical power demands for the five investigated hot standby variants is provided in Table IV. The table lists the individual fuel flows as their equivalent chemical powers ($LHV_{\text{fuel}}\cdot\dot{m}_{\text{fuel},i}$) for the main anode supply line, the afterburner, and, where applicable, the cathode pre-heating burner, together with the total fuel demand. In parallel, the electrical power demand ($P_{\text{el},i}$) is given as the sum of the power requirements for all blowers and, if applicable, the cathode electric heater, as well with the total electrical power demand.

In the rightmost column of Table IV, the combined fuel demand per module ($P^{\text{mod}}_{\text{comb,standby}}$) is presented as the main performance indicator for each variant. The value is calculated by converting the module's electrical power demand into its fuel equivalent, assuming that the required electricity is provided by another active module within the same unit operating at its maximum efficiency (63.4%, see Table I):

$$P^{\text{mod}}_{\text{comb,standby}} = LHV_{\text{fuel}}\cdot\dot{m}_{\text{fuel,standby}} + P^{\text{mod}}_{\text{el,standby},k} = LHV_{\text{fuel}}\cdot\sum_i \dot{m}_{\text{fuel},i} + \frac{1}{\eta_{\text{mod,max}}}\sum_i P_{\text{el},i} \quad [6]$$

As all hot standby variants share identical anode-side boundary conditions regarding autothermal reforming operation and anode recirculation, the base fuel supply to the anode loop remains constant at 0.04 g s^{-1}, corresponding to an equivalent chemical power input of approximately 2 kW_{chem} per module.

For Variant 1, a considerable additional fuel flow of 11.6 kW_{chem} must be combusted in the afterburner to reach the target cathode inlet temperature of 700 °C. To achieve this, the afterburner temperature must be elevated to approximately 900 °C, approaching the upper material limits of the component. Owing to the low process mass flows, the combined electrical demand for the air and anode recirculation blowers is only about 0.1 kW_{el}, resulting in a total combined fuel demand of 13.8 kW_{chem}.

In Variants 2 and 3, the use of a dedicated cathode pre-heating component considerably reduces the fuel requirement for maintaining a stable afterburner temperature. Both configurations show similar total energy demands of approximately 6.5 kW, either as electric power (Variant 2) or as chemical energy in the cathode burner (Variant 3). When accounting for the conversion efficiency penalty associated with generating electricity in an active module, Variant 2 yields the highest combined demand (15.3 kW_{chem}), whereas the fuel-fired cathode burner in Variant 3 requires only 11.6 kW_{chem}.

When COGR is implemented (Variants 4 and 5), a significant improvement in overall efficiency is achieved, reducing the combined fuel demand to 6.5 kW_{chem} and 5.2 kW_{chem}, respectively. This improvement can be attributed to three main factors: (i) enhanced heat integration efficiency, as the heater or burner needs only to compensate for the stack and COGR line heat losses, (ii) reduced blower power consumption, because only about 15% of the total cathode air mass flow enters and exits the recirculation loop at a recirculation ratio of 85% (see Table I), and (iii) no additional fuel demand at the afterburner, as the decreased external air flow leads to higher exhaust gas temperatures around 810 °C.

Discussion of results and translation to unit performance.—The overall unit part-load efficiency was derived by inserting the module-level fuel and power demand results from Table IV into the efficiency formulation given in Eq. 2. The resulting efficiencies are presented in Fig. 6, where discrete markers represent the achievable operating points of a unit comprising nine modules. The dashed lines serve as visual fits to indicate the overall trend.

Table IV. Electrical, fuel and combined power demand in hot standby.

	Fuel demand in kW_{chem}				Electrical power demand in kW_{el}			Combined in kW_{chem}
Var.	Anode	After-burner	Cathode Burner	Total	BoP Blowers	Cathode El. Heater	Total Electr.	
1	2.00	11.6	—	**13.6**	0.11	—	**0.11**	**13.8**
2	2.00	2.94	—	**4.94**	0.11	6.47	**6.58**	**15.3**
3	2.00	2.82	6.64	**11.5**	0.11	—	**0.11**	**11.6**
4	2.00	—	—	**2.00**	0.04	2.84	**2.88**	**6.53**
5	2.00	—	3.17	**5.17**	0.04	—	**0.04**	**5.23**

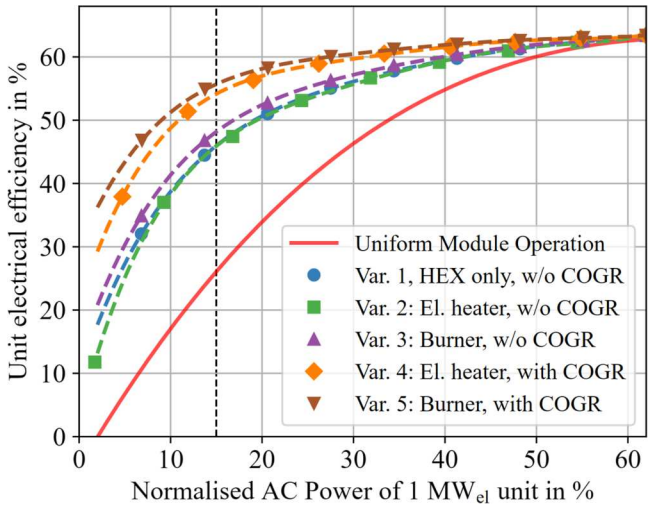


Figure 6. Low part load unit efficiencies using hot standby Variants 1 to 5, compared to uniform module operation. The low part-load demand spike at 15% rated power from Fig. 1b is highlighted as black dashed line.

The unit efficiency curves corresponding to the hot standby variants without COGR (Variants 1–3) are closely aligned and already lie well above the solid reference line representing uniform module operation. At the part-load point of 15% of rated power, which was identified to be of particular relevance for maritime operation in Fig. 1, the uniform operation yields an electrical efficiency of only 26%. In contrast, implementation of hot standby operation for Variants 1–3 increases the achievable efficiency to values between 45.9% and 48.2%.

A distinct further improvement is observed for the configurations including COGR (Variants 4 and 5), reaching unit efficiencies of 54.1% and 55.7% at 15% load, respectively. These results correspond to more than a twofold efficiency increase compared to uniform operation and an additional gain of 7–9 percentage points over the variants without COGR. These findings highlight the pronounced advantage of incorporating COGR in SOFC systems expected to experience extended hot standby periods. While a previous study identified only limited gains in electrical efficiency during active operation for systems employing COGR,[3] the present results demonstrate that the concept is particularly beneficial under long-duration standby conditions for modular systems, theoretically enabling high efficiencies in drivetrains exceeding 50% even at power demands as low as 10% of rated AC output.

Configurations employing a cathode pre-heating burner exhibit the best respective overall performance. At the 15% load point, these configurations achieve efficiencies 2.3 percentage points higher than the corresponding electric-heater variants without COGR (Variant 3 vs 2) and 1.5 percentage points higher with COGR (Variant 5 vs 4). Despite this efficiency advantage, the use of combustion-based pre-heating introduces combustion products into the cathode gas stream, which may accelerate degradation of the cathode material. Simulation results indicate that for Variant 3 (without COGR), the cathode inlet gas contains approximately 1.0 mol-% H_2O and 0.5 mol-% CO_2, whereas for Variant 5 (with 85% COGR), the recirculated flow increases these values to 3.2 mol-% H_2O and 1.6 mol-% CO_2, respectively. Such concentrations, along with other combustion particles, can promote degradation mechanisms such as strontium segregation and carbonate formation, the severity of which depends strongly on the specific cathode material and its fabrication. Consequently, while the system-level results confirm the thermodynamic advantages of burner-based pre-heating, future SOFC system designs should carefully consider the trade-off between these modest efficiency gains and potential long-term degradation risks associated with cathode exposure to combustion products.

Sensitivity study on assumed variables.—Following the assessment of the different heat-integration concepts under comparable operating conditions, the underlying model assumptions are now revisited to evaluate the uncertainty of the obtained results and to identify potential pathways for further power demand reduction. This analysis also provides insight into the operational constraints that may arise from varying system design or boundary parameters.

The sensitivity study was conducted for the best-performing configuration (Variant 5), featuring COGR and a cathode pre-heating burner. The results are illustrated in Fig. 7, which shows the influence of selected parameters on the combined fuel demand per module $P^{\text{mod}}_{\text{comb,standby}}$. The reference condition in Fig. 7 is marked as the red dashed vertical line, representing the combined fuel demand of 5.23 kW_{chem} as reported in Table IV. Blue horizontal bars indicate the magnitude and direction of the change in combined fuel demand relative to variations in the respective parameters.

Five parameters were investigated, starting with those associated with higher uncertainty, overall pressure losses and BoP heat losses, followed by adjustable operating variables: stack inlet temperature, cathode recirculation ratio, and cathode mass flow.

Pressure loss.—The pressure drops in all modelled components were varied by applying the scaling factor ϵ in Eq. 5, covering a range of three orders of magnitude ($0.1 < \epsilon < 10$). While rated-pressure design values are generally constrained by the SOFC stacks' allowable differential pressure to ambient, the linear dependency assumption in Eq. 5 introduces uncertainty under low-flow conditions. Nevertheless, the analysis shows that even a tenfold increase in total pressure losses would raise the combined fuel demand by only about 0.3 kW_{chem}, demonstrating a low sensitivity to this parameter and blower power in general.

BoP heat losses.—The total BoP heat loss was varied by ±0.7 kW_{th} from the reference assumption, while the stack heat losses, based on experimental data, were kept constant. The combined fuel demand changes by approximately 0.4 kW_{chem} for this variation. However, BoP heat losses exceeding roughly 1.0 kW_{th} lead to pre-reformer outlet temperatures dropping below the defined minimum of 700 °C, as indicated by the hatched zone in Fig. 7, which would impact the degree of pre-reforming significantly.

Cathode inlet temperature.—In all prior simulations, the cathode inlet temperature was maintained at 700 °C, corresponding to the typical average operating temperature of anode-supported SOFC stacks, enabling rapid return to power-generating mode. For extended standby periods such as harbor stays, lowering the standby temperature could reduce heat losses and save fuel at the expense of longer reheating times. When the stack temperature is decreased by 100 K to 600 °C, the combined fuel demand theoretically decreases by up to 0.5 kW_{chem} per module. However, due to the strong thermal coupling within the anode loop, lowering the stack temperature also reduces the pre-reformer temperature unless counteracted by increased fuel and air supply to the autothermal reformer (not applied here). Similar to the heat losses, this causes a reduction in methane conversion, promoting endothermic internal reforming within the stacks during standby. Thus, despite its fuel-saving potential, deliberate temperature reduction is not recommended for the configurations studied.

Cathode recirculation ratio.—At a constant cathode mass flow (10% of rated operation), the recirculation ratio RR_{COGR} was varied around the nominal value of 0.85. Increasing this ratio to 0.95 reduces the combined fuel demand by approximately 0.9 kW_{chem}, as less external air needs to be heated. However, beyond a threshold of 0.92, the air supply to the afterburner becomes insufficient to maintain material-safe temperatures, causing the afterburner temperature to exceed 900 °C, defining a practical upper limit for the recirculation ratio.

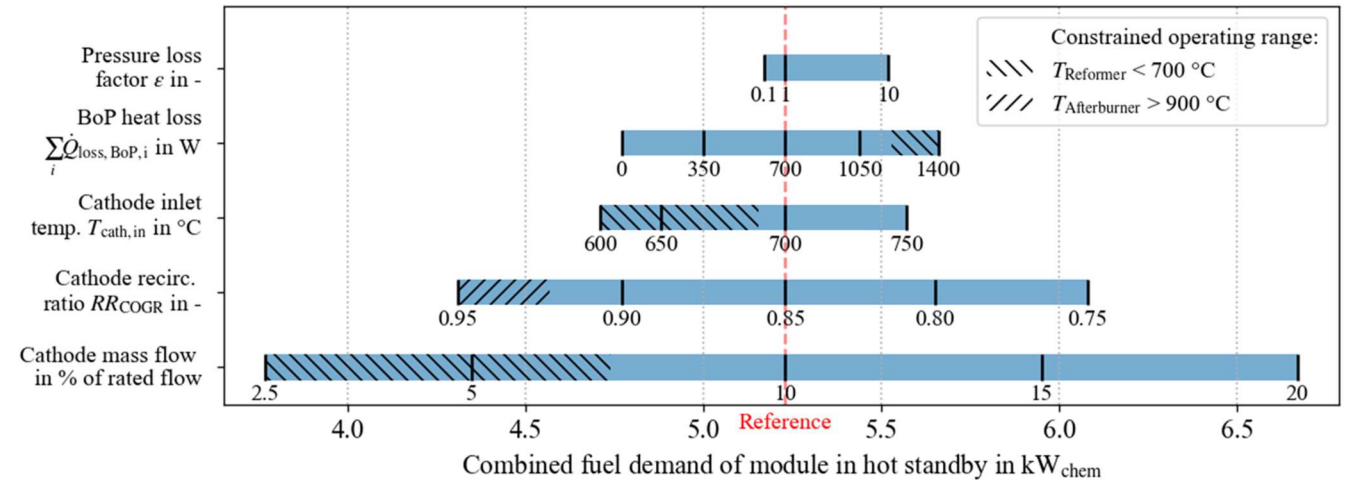


Figure 7. Sensitivity study for Variant 5 regarding unknown and remaining controllable parameters for hot standby.

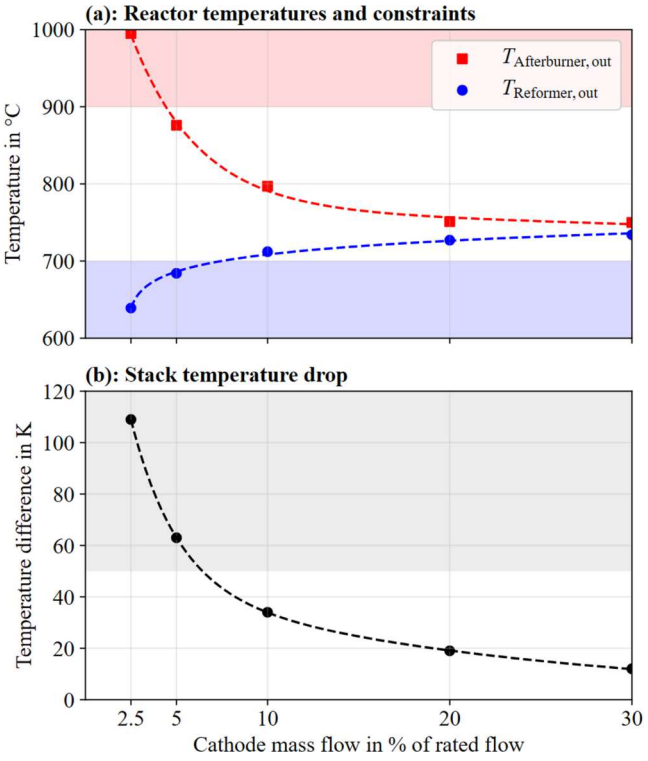


Figure 8. Impact of cathode mass flow variation on (a) afterburner and pre-reformer temperatures and (b) drop of stack temperature between cathode inlet and outlet.

Cathode mass flow.—Varying the total cathode mass flow at a constant $RR_{COGR} = 0.85$ produced the strongest influence among all investigated parameters. In theory, reducing the cathode mass flow could lower the combined fuel demand by more than 1 kW_{chem}. The underlying mechanism is a decrease in cathode preheating demand with reduced mass flow. However, lower air mass flow also diminishes the heat capacity flow of the cathode stream and reduces the effectiveness of downstream heat exchangers and heat transfer to pre-reformer, where fuel flow remains nearly constant. As shown in Fig. 8, both the stack inlet-outlet temperature difference and the pre-reformer temperature drop sharply at significantly reduced cathode mass flows, while the afterburner temperature simultaneously rises. Reducing the cathode mass flow below approximately 7.5% of the rated value would therefore cause both the pre-reformer and stack temperatures to fall below their respective lower limits. Realistically, this allows only about 0.5 kW_{chem} of additional fuel savings.

The single-parameter sensitivity analysis demonstrates that modest optimization around the reference operating point is possible, but the chosen baseline parameters already represent a well-balanced operating condition. Further deviations could lead to unfavorable trade-offs such as excessive temperature gradients, material stress, or increased risk of pre-reformer and stack degradation. For future design of larger systems with multiple modules, multi-parameter or Monte Carlo approaches could be carried out to capture an optimal hot standby performance with small but reliable design margins.

Impact of module number on unit part load efficiency.—The analyses presented so far (Figs. 4 and 6) considered only discrete load points resulting from configurations in which the active modules operate at their individual maximum efficiency points. For the reference case of a unit containing nine modules, this results in nine discrete points below the maximum-efficiency operating point at 62% of rated power. A remaining question is how the active modules should be operated between these discrete points to achieve the highest possible unit efficiencies, and how the overall number of installed modules N influences this behavior. To address these questions, a first optimization study was conducted assuming a uniform actuation strategy, where all active modules operate at the same relative load point, defined as $x_{act} = P^{mod}_{el,act}/P^{mod}_{el,rated}$ to deliver a total unit output power of $P^{unit}_{el} = |\sum_{k=1}^{N_{act}} P^{mod}_{el,act,k}| - \sum_{k=1}^{N-N_{act}} P^{mod}_{el,standby,k}$. The module efficiency as a function of load, $\eta^{mod}_{el,LHV,AC}(x_{act})$, is implemented as a fifth-order polynomial fit derived from detailed system simulations (shown as the red curves in Figs. 1 and 4). The coefficients for this fit are provided in the appendix. With the module's operating point x_{act} and the number of active modules N_{act} as free variables, an optimization problem is formulated using Eq. 2 to determine the combination that maximizes unit efficiency across the full load range:

$$\max_{x_{act},\ N_{act}} \left(\frac{\left| \sum_{k=1}^{N_{act}} P^{mod}_{el,act,k} \right| - \sum_{k=1}^{N-N_{act}} P^{mod}_{el,standby,k}}{LHV_{fuel} \cdot \left(\sum_{k=1}^{N_{act}} \dot{m}_{fuel,act,k} + \sum_{k=1}^{N-N_{act}} \dot{m}_{fuel,standby,k} \right)} \right). \quad [7]$$

For this assessment, the fuel and power hot standby demands of the most efficient configuration, Variant 5 (as listed in Table IV), were used. The optimization results are depicted in Fig. 9. The left panel (Fig. 9a)

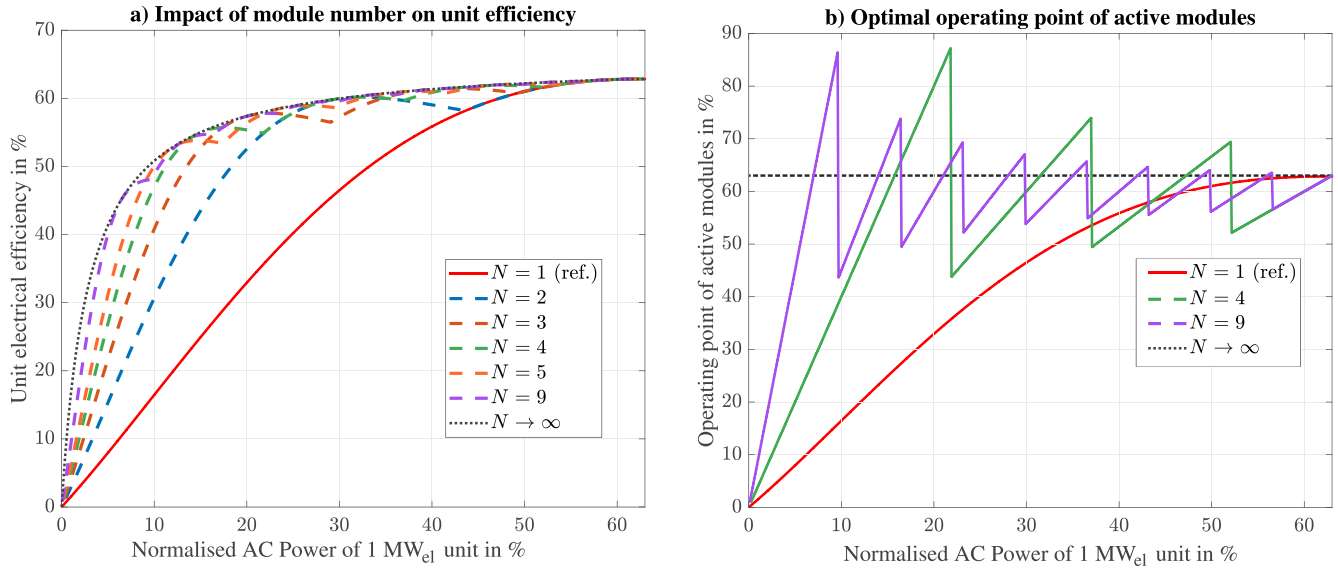


Figure 9. Variation of number of modules N within the overall powertrain. (a) Impact on unit efficiency over load range. (b): associated optimized uniform part load actuation of active modules to achieve the respective maximum efficiencies.

shows the achievable unit efficiency over the load range for varying the total module numbers N. The red solid line represents the limiting case of a single-module unit ($N = 1$), while increasing the number of modules progressively raises efficiency and smooths the efficiency curve. As N approaches infinity, the results converge toward a maximum theoretical efficiency limit, shown as the dotted black curve.

Figure 9b illustrates the corresponding actuation strategy of the active modules by plotting the module operating point x_{act} as a function of unit load for two representative configurations with four (green) and nine (purple) modules. As load increases, the optimizer gradually raises the operating point of the active modules until reaching a threshold above the individual module's maximum-efficiency point. At this point, the inclusion of an additional module from hot standby and a simultaneous reduction of load on all active modules yields a higher overall unit efficiency. This transition is visible as a local change in the slope of the efficiency curves in Fig. 9a, where the trend shifts from decreasing to increasing efficiency.

For the uniform actuation strategy considered, the magnitude of these step-like load changes diminishes both with increasing unit load and with a larger total number of modules, indicating smoother operation for more finely discretized multi-modular systems. Even a unit consisting of only four modules, however, demonstrates a substantial efficiency advantage at low part-load operation compared to uniform full-module operation, highlighting the inherent benefit of a modular SOFC powertrain with efficient hot standby capability for future full-SOFC ship applications.

Extending this concept beyond the scale of a single unit (fuel-cell room), multiple MW units could be controlled collectively as one system. The resulting high number of individually actuatable modules would further smooth the efficiency curve and enhance load-following capability. In practice, advanced rule-based or predictive control strategies could achieve comparable efficiencies while reducing the frequency of module switching between active and standby states, thus improving system robustness and lifetime.

It should be noted that the present analysis is limited to steady-state conditions and does not account for transient effects during transitions into and out of hot standby. Future investigations should include these dynamic behaviors to assess their impact on overall system efficiency and operational flexibility. The findings presented here motivate experimental validation of both (i) the proposed hot standby variants at module level and (ii) the efficiency-optimized multi-modular operation strategy under dynamic load profiles representative of real ship operation.

Conclusions

To the authors' knowledge, this work presented the first systematic assessment and quantification of hot standby power demands for modular SOFC systems and its impact on marine powertrain applications. By combining thermodynamic process system simulations of an SOFC module developed in the Nautilus project with realistic ship boundary conditions, the study demonstrated that efficient hot standby operation on module level is essential for maintaining high powertrain efficiencies at low loads, a key requirement for SOFC ship integration scenarios where the SOFC would be the primary energy supplier for propulsion, auxiliary and hotel load.

A natural-gas fed SOFC module, consisting of a hot box with six stacks with anode-supported cells, high-temperature anode recirculation and pre-reforming was taken as the baseline model for a thermodynamic simulation study. Five hot standby variants were investigated, all employing autothermal reforming but differing in their heat integration concept and amount of additionally needed hot standby components. Simulation results revealed that configurations employing cathode off-gas recirculation (COGR) offer the lowest overall standby energy demand, with equivalent fuel requirements as low as 5.2 to 6.5 kW$_{chem}$ for a 112 kW$_{el}$-rated module. When extrapolated to a multi-modular level, such variants could more than double the achievable efficiency in the low part load regime compared to the module's standalone efficiency curve.

A sensitivity analysis was subsequently performed for the most efficient hot standby configuration, combining COGR with a cathode-side burner. Among all investigated parameters, the cathode mass flow exhibited the strongest influence on the combined fuel demand, but it also governs the temperature drop within the stacks and can lead to temperature limit violations in the pre-reformer and afterburner if reduced excessively.

Furthermore, while a larger number of modules in the overall powertrain enables smoother actuation and enables consistently high part-load efficiencies, the results show that even a system with as few as four modules can already achieve a substantial improvement in low-load efficiency compared to uniform operation.

The results highlight that hot standby operation should be explicitly considered in the design and sizing process of future maritime SOFC systems and that experimental validation of the most efficient hot standby concepts should be carried out.

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Appendix

The module efficiency curve in Figs. 1, 4 and 6, indicated as red solid line and generated from system simulation using the identical modeling framework, is given as third-order polynomial fit, where P is the module net AC power referred to the rated module power. Both power and efficiency are given in %:

$$\eta_{\mathrm{mod}}(P) = -4.8909 + 2.4661 \cdot P - 0.0282 \cdot P^2 + 9.780 \cdot 10^{-5} \cdot P^3$$

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